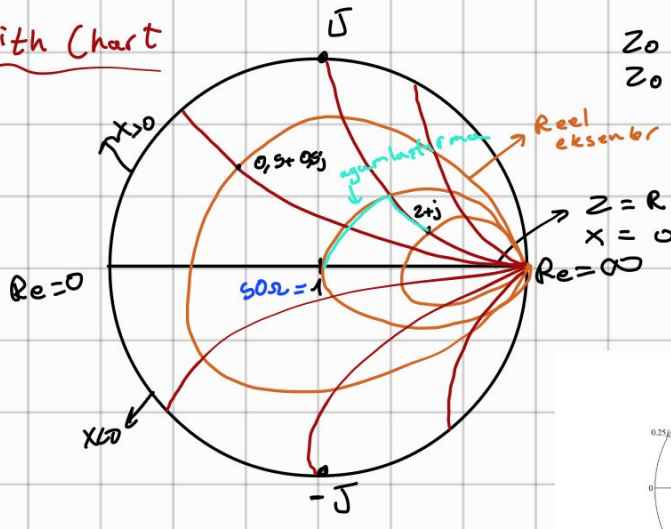
Ders 4

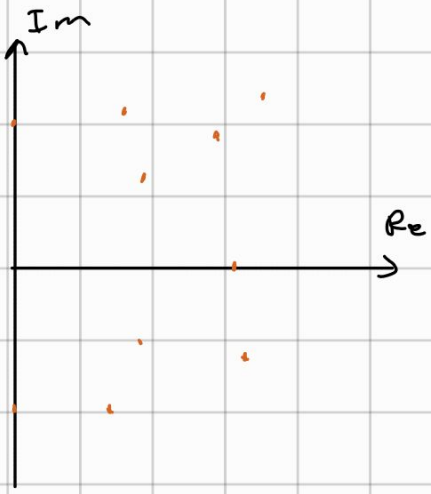
Smith Chart



$Z = R + jX$

Z_0 normalize
$Z_0 = 50\Omega$

Empedans



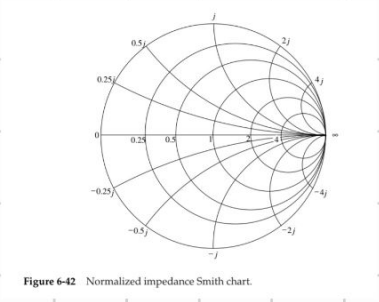


Figure 6-42 Normalized impedance Smith chart.

$Z_{in} = R + j\underbrace{\omega X_{im}}_{X}$

- Impedance matching
- Finding transmission line length
- Finding the value of L or C used for matching
- Finding reflection coefficient of both load or sour
- Convertin Z to admittance Y
- Finding VSWR (Voltage Standing wave ratio)

Softwares

çip düzeyinde tasarım

1 - (keysight) ADS Advance Design System
EM simulation

2 - (Cadence) NI AWR

3 - CST (Dassault) Özellikle anten tasarımcıları için

4 - HFSS (Ansys)

Measurement Devices

1) Spectrum Analyzer: frekanslar; güçlerini ---
(Frequency Range) (10 MHz-18GHz)

2) Signal Analyser:

Spectrum Analyseri ilaveten demodulation yapılabilir

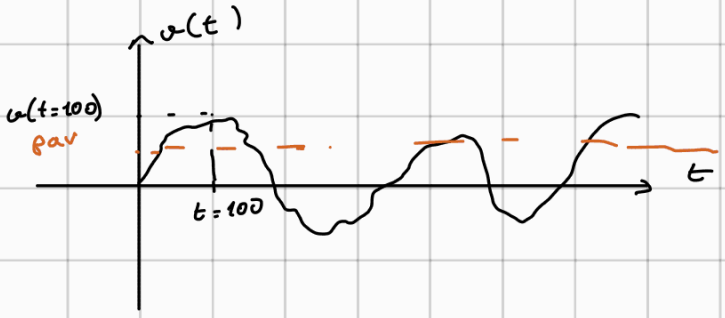
3) Network Analyzer :

S param $\rightarrow$ ρ-domain or Smith chart

4) Signal generator : yüksek frekanslı singal üretir

5) Microwave / RF power Meter

Instantaneous Power (anlık güç)



$$p(t) = v(t)\,i(t) = \frac{v^2(t)}{R}$$

Bizim için ortalama güç önemli

$$P_{ave} = \frac{1}{T}\int_0^T p(t)\,dt = \frac{1}{R}\,\frac{1}{T}\int_0^T v^2(t)\,dt$$

$$\langle v^2(t)\rangle = \frac{1}{T}\int_0^T v^2(t)\,dt \qquad v^2(t)\text{'nin ortalaması}$$

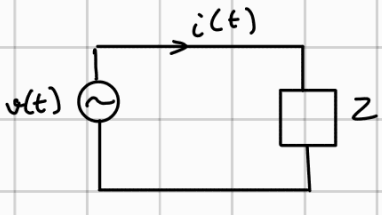
$$\sqrt{\langle v^2(t)\rangle} = V_{rms} \text{ (root mean square)}$$

$$v(t) = \frac{2}{A}\cos(\omega t)$$

$$\langle V^2(t)\rangle = \frac{1}{T}\int_0^T v^2(t)\,dt = \frac{A^2}{2}$$

$$R = 50\,\Omega \Rightarrow P_{avg} = \frac{\langle V^2(t)\rangle}{R} = \frac{A^2}{2\cdot 50} = \frac{1}{25}\ W$$

Power & Phasor



$$V = |V|e^{j\theta_V} \rightarrow v(t) = |V|\cos(\omega t + \theta_V)$$

$$I = |I|e^{j\theta_I} \rightarrow i(t) = |I|\cos(\omega t + \theta_I)$$

$$z = \frac{V}{I} = \frac{|V|e^{j\theta_V}}{|I|e^{j\theta_I}} = \frac{|V|}{I}\,e^{j(\theta_V - \theta_I)}$$

$Z = R \Rightarrow I = \frac{V}{Z} = \frac{|V| e^{j\theta_V}}{R} = \frac{|V|}{R} e^{j\theta_V}$

$Z = jwL = wL \cdot e^{j90} \Rightarrow I = \frac{V}{Z} = \frac{|V| e^{j\theta_V}}{wL e^{j90}} = \frac{|V|}{wL} e^{j(\theta_V - 90)}$

akım gerilime göre 90° derece geridedir



$p(t) = v(t) \cdot i(t)$

$= |V||I| \cos(wt + \theta_V) \cos(wt + \theta_I)$

$= \frac{|V||I|}{2} \left[\cos(2wt + \theta_V + \theta_I) + \cos(\theta_V - \theta_I) \right]$

$P_{avg} = \frac{1}{T} \int_0^T p(t)\,dt = \frac{|V||I|}{2} \cos(\theta_V - \theta_I)$

$= \frac{|V|}{\sqrt{2}} \frac{|I|}{\sqrt{2}} \cos(\phi) \qquad \phi \triangleq \theta_V - \theta_I$

$$P_{ave} = V_{rms} I_{rms} \cdot \cos(\phi)$$

$Z = L \rightarrow P_{ave} = 0$ → Nasıl olabilir (reaktif güç var)

$Z = R \rightarrow P_{ave} = V_{RMS} \cdot I_{RMS}$

Complex Power

$V = |V| e^{j\theta_V}$ $\qquad V_{rms} = \frac{|V|}{\sqrt{2}} e^{j\theta_V}$

$I = |I| e^{j\theta_I}$ $\qquad I_{rms} = \frac{|I|}{\sqrt{2}} e^{j\theta_I}$

$S = P + jQ$

$S = \frac{1}{2}(V \cdot I^*)$ or $(V_{rms} \cdot I_{rms}^*)$

$= \frac{1}{2}(|V| e^{j\theta_V} \cdot |I| e^{-j\theta_I}) = \frac{1}{2}|V||I| e^{j(\theta_V - \theta_I)}$

$= \underbrace{\frac{1}{2}|V||I| \cos(\theta_V - \theta_I)}_{P} + j \underbrace{\frac{1}{2}|V||I| \sin(\theta_V - \theta_I)}_{Q}$

P: average power or active Power $P = Real\{S\}$ [watt-W]

Q: reactive power $Q = Imag\{S\}$ [VAR] Voltage Amper Reactive

	Z	P	Q
Resistor	R $\theta_v = \theta_I = 0$	$P = \frac{\lvert V^2 \rvert}{R} > 0$ absorbs power	0
Inductor	$j\omega L$ $\theta_v - \theta_I = 90^\circ$	0	$j\frac{\lvert V \rvert^2}{\omega L} > 0$ absorbs power (VAR)
Capacitor	$\frac{1}{j\omega C}$ $\theta_v - \theta_I = -90^\circ$	0	$-\frac{1}{j}\, j\lvert V \rvert^2 \omega C < 0$ generates VAR

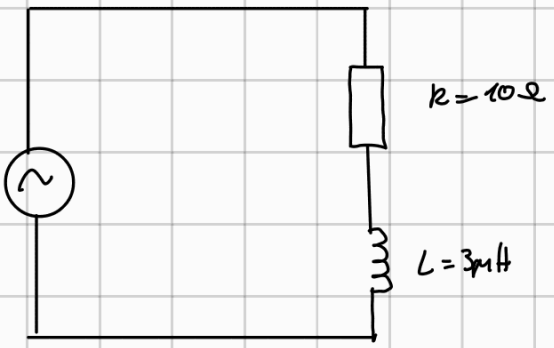
HW ①

1) $v(t) = 2\cos(\omega t + 30)$

$Z = 5 + j10$

Find I and Pave

2)

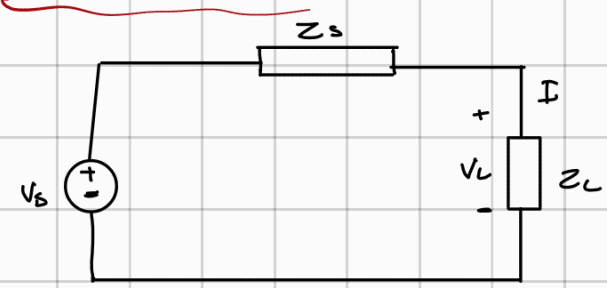


$v(t) = 2\cos(\omega t)$

$f = 1MHz$

Find S = ?

Maximum Power



$$P_{ave(load)} = P_L = \frac{1}{2} Re\{V_L I_L^*\} = \frac{1}{2} Re\left\{\frac{Z_L}{Z_s + Z_L} V_s \left(\frac{V_s}{Z_s + Z_L}\right)^*\right\}$$

(V_L = $\frac{Z_L}{Z_s + Z_L} V_s$)

$$= \frac{1}{2} Re\left\{V_s \cdot V_s^* \cdot \frac{Z_L}{Z_s + Z_L} \frac{1}{Z_s^* + Z_L^*}\right\}$$

$$= \frac{1}{2} \frac{|V_s|^2}{|Z_s + Z_L|^2} \cdot \mathrm{Re}\{Z_L\}$$ → $R_L + X_L$

$$P_L = \frac{1}{2} \frac{|V_s|^2}{|R_s + jX_s + R_L + jX_L|^2} \cdot R_L$$

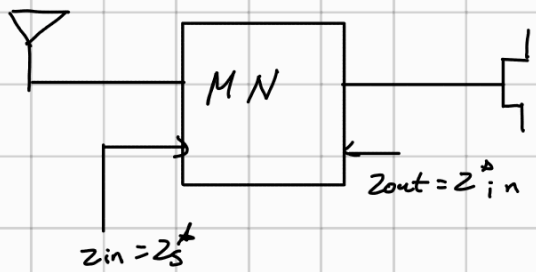
maximize etmek için $Z_s = Z_L^*$

$$Z_s = Z_L^* \Rightarrow P_L = \frac{1}{2} \frac{|V_s|^2}{|R_s + jX_s + R_s - jX_s|} \cdot R_s$$

$$= \frac{1}{2} \frac{|V_s|^2}{|2R_s|^2} \cdot R_s$$

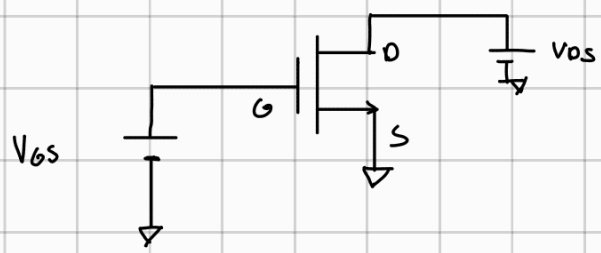
$$P_L = \frac{1}{8} \frac{|V_s|^2}{R_s}$$

yükteki max power kaynağa bağlı

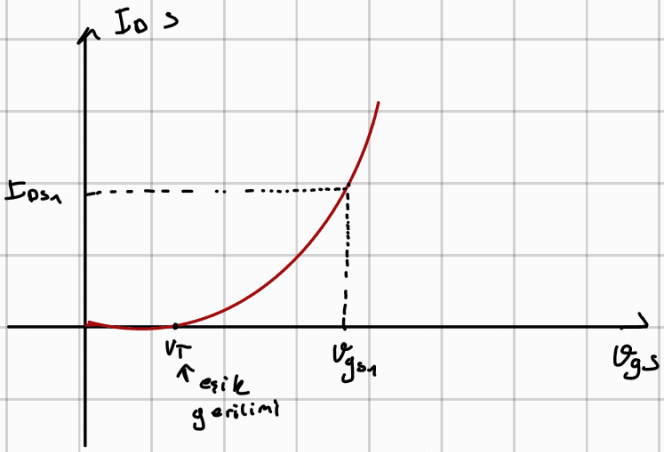


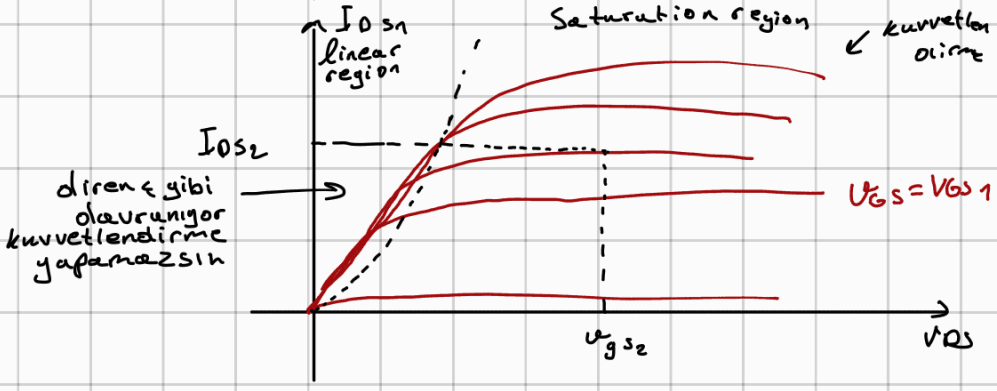
kuvvetlendirciler

MOS transistörler

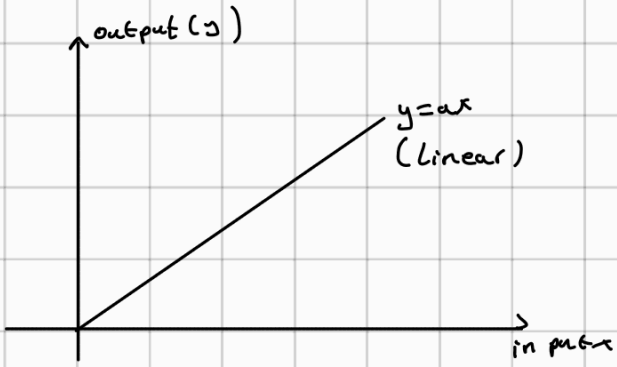


$$I_D = k \frac{W}{L} (V_{GS} - V_T)^2$$ ← set





Linearity

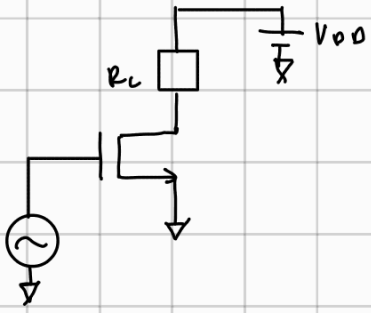


$$y = ax + bx^2 + cx^3 + \cdots$$

↑ non linear

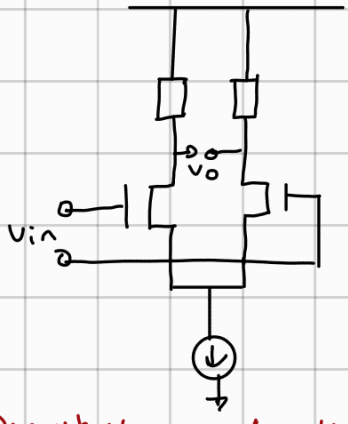
$x \ll$ (to choose) $\Rightarrow bx^2 + cx^3 \cong 0$

Ex

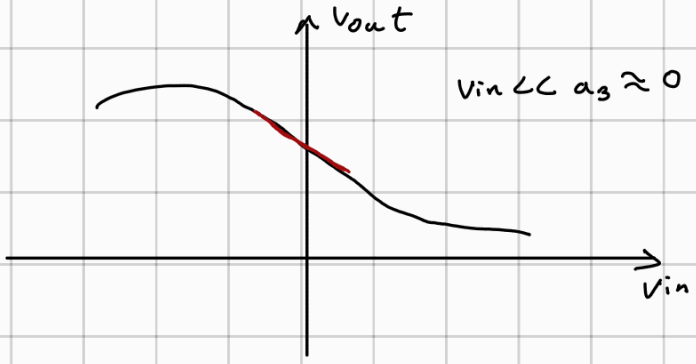


$$V_{out} = V_{DD} - k\frac{W}{L}(V_{GS}-V_T)^2 \cdot R_L$$
$$= DC + a_2 V_{GS}^2$$

Differential Pair



$$V_{out} = -\sqrt{2k\frac{W}{L} I_D} \cdot R_D \cdot V_{in} + \frac{2k^{3/2}}{\delta\sqrt{I_D}} R_D V_{in}^3$$
$$= a_1 \cdot V_{in} + a_3 V_{in}^3$$

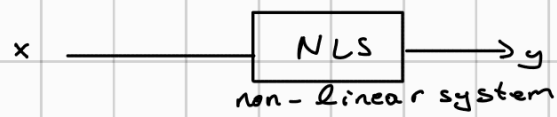


$V_{in} \ll$ (compared to) $a_3 \approx 0$

Drawbacks of Nonlinearity

① Harmonic Distortion

$$y = a_1 x + a_2 x^2 + a_3 x^3$$



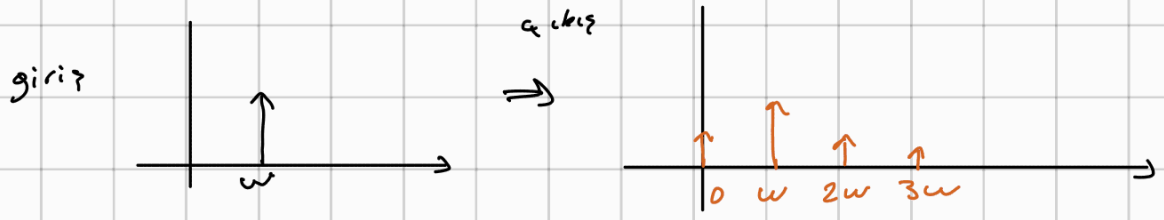
$x(t) = A\cos\omega t$

$y(t) = a_1 A\cos\omega t + a_2 (A\cos\omega t)^2 + a_3 (A\cos\omega t)^3$

$(A\cos\omega t)^2 = \frac{A^2}{2}(1+\cos 2\omega t)$

$(A\cos\omega t)^3 = \frac{A^3}{4}(3\cdot\cos\omega t + \cos 3\omega t)$

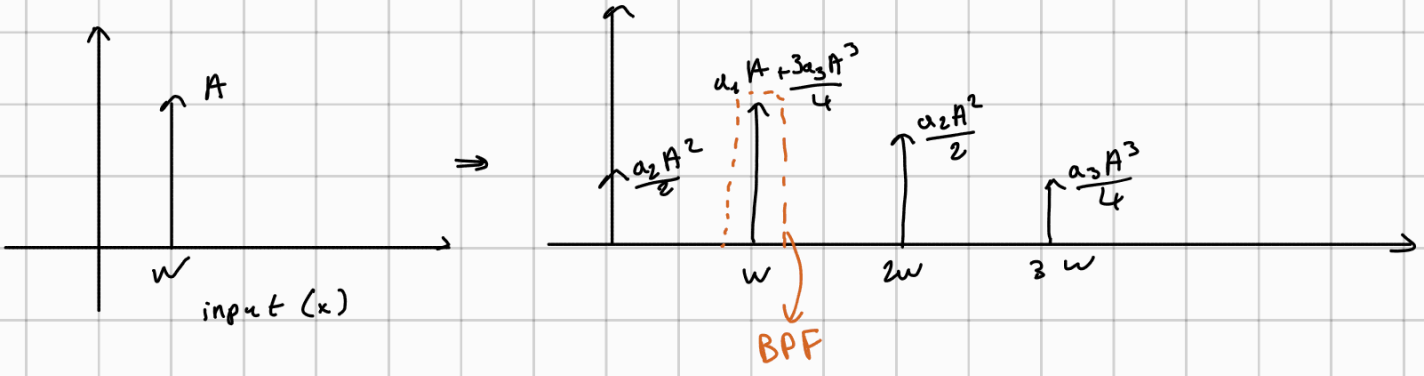
$y(t) = a_1 A\cos\omega t + a_2 \frac{A^2}{2}(1+\cos(2\omega t)) + a_3 \frac{A^3}{4}(3\cos\omega t + \cos 3\omega t)$

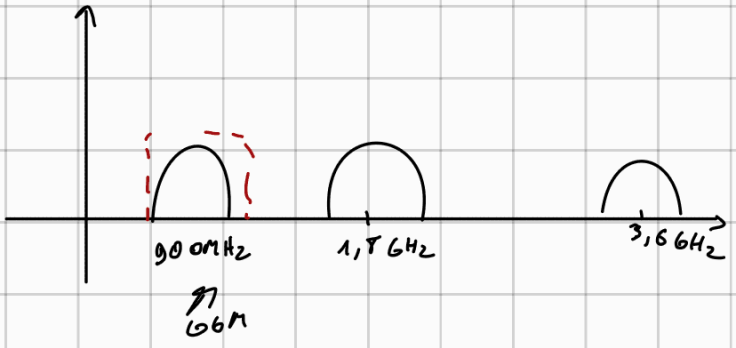


$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos\omega t + \frac{a_2 A^2}{2}\cos 2\omega t + \frac{a_3 A^3}{4}\cos 3\omega t$

2nd harmonic

3rd





(2) Gain compression

$$V_{out} = \underbrace{-\sqrt{2k\frac{W}{L}I_D}\,R_D V_{in}}_{a_1} + \frac{2k^{3/2}}{8\sqrt{I_D}} R_D V_{in}^3$$

a_1, a_3 have different sign

Most of the time $\boxed{a_1 \cdot a_3 < 0}$

$$\text{Gain} = \frac{V_{out}(\text{fundamental})}{V_{in}} = \frac{a_1 A + \frac{3a_3 A^3}{4}}{A}$$

$Gain = a_1 + \frac{3}{4} a_3 A^2$

1) $a_1 > 0$, $a_3 < 0 \rightarrow Gain = a_1 - \frac{3}{4} |a_3| A^2$ → küçülür

2) $G_{in} \propto A^2$

girişin genliği çok yüksekse kazanç düşer.